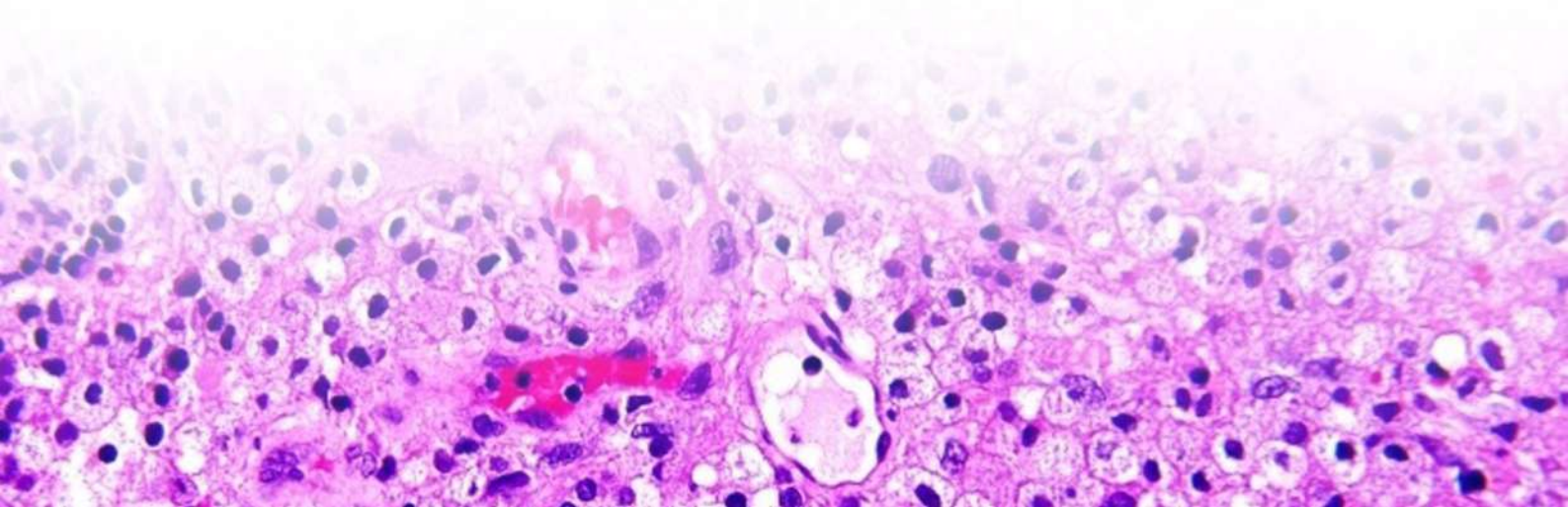
Student Notes

PATHOLOGY

1st Edition

Unit 20

Female Genital Tract



Preface

Agam is a group of budding medicos, who are currently doing their under graduation in various Medical Colleges across Tamil Nadu and Pondicherry. The group was initiated on 18th November 2017, in the vision of uniting medicos for various social and professional causes.

We feel delighted to present you Agam Pathology notes prepared by Agam Divide and Rule 2020 Team to guide our fellow medicos to prepare for university examinations.

This is a reference work of 2017 batch medical students from various colleges. The team took effort to refer many books and make them into simple notes. We are not the authors of the following work. The images used in the documents are not copyrighted by us and is obtained from various sources.

Dear readers, we request you to use this material as a reference note, or revision note, or recall notes. Please do not learn the topics for the 1st time from this material, as this contain just the required points, for revision.

Acknowledgement

On behalf of the team, Agam would like to thank all the doctors who taught us Pathology. Agam would like to whole heartedly appreciate and thank everyone who contributed towards the making of this material. A special thanks to Vignesh M, who took the responsibility of leading the team. The following are the name list of the team who worked together, to bring out the material in good form.

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THE FEMALE GENITAL SYSTEM

ESSAY:

1. Serous tumours
2. Germ cell ovarian tumours
3. Surface epithelial tumours of ovary
4. Classify ovarian neoplasms. Molecular pathogenesis and morphology of ovarian tumours
5. Cervical cancer

SHORT NOTES

1. Choriostoma
2. Brenner's tumour
3. Benign cystic tetatoma
4. Endometriosis
5. Malignant tumours of endometrium
6. Choriocarcinoma
7. Cervical intraepithelial neoplasia
8. Hydatiform mole
9. Cervical cancer – Lab diagnosis
10. Uterine leiomyoma
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SHORT ANSWERS:

1. Sciller Duvel bodies
2. Call Exner bodies
3. Gross appearance of mature cystic teratoma
4. Histological features of dysgerminoma
5. Classification of gestational trophoblastic disease
6. Adenomyosis
7. Dermoid cyst of ovary
8. Microscopic features of Leiomyoma
9. Alpha feto protein

UPDATES

ESSAY

1. SEROUS TUMORS

Incidence:

- Serous tumors account for about 30% of all ovarian tumors and about over 50% of ovarian epithelial tumors.
- About 70% are benign or borderline. About 30% are malignant.

Age Group

- Benign and borderline tumors: Most common between 20 and 45 years.
- Serous carcinomas: Occur later in life.

GROSS

Benign serous cystadenoma -

- **Size:** Varies and may measure from 15 to 30 cm in diameter.
- **Appearance:** It contains one or more thin-walled cysts.
- The cyst wall is smooth glistening without any epithelial thickening.
- **Content:** Lumen filled with clear watery serous fluid.
- **Cut section**
  - Shows smooth internal surface and small papillae projecting into the cavity.
  - Rarely papillae seen projecting from the outer surface of ovary (surface papilloma).
- Bilaterality is common and found in about 20% of benign serous cystadenomas.

Borderline tumors

- Vary in size, usually >5 cm and typically cystic.
- Cut section shows cystic cavities filled by increased number of dense and closely packed, cauliflower-like papillary projections.
- Bilaterality: 30%.

Malignant serous adenocarcinoma

- Usually large.
- Shows a mixture of solid and cystic areas with large solid or papillary areas.
- Other features of malignancy includes: tumor irregularity, and fixation or nodularity of the capsule.

MICROSCOPY

Benign serous tumors:

- Cysts are lined by non-stratified or stratified cuboidal to columnar epithelial cells (similar to lining epithelium of normal fallopian tube).
- Cilia are present though sometimes only focally. May also show:
  - **Papillae with a fibrovascular core:** They are covered by a single layer of epithelium similar to that of the cyst lining.
  - **Psammoma bodies:** They are dystrophic calcified tumor cells.

Borderline tumors:

- They are non-invasive tumors with greater epithelial proliferation and cytological atypia than benign but less than low-grade serous carcinoma.
- Stratification of the epithelium (multilayering)
- The epithelial proliferation may produce a delicate, papillae pattern termed as "micropapillary carcinoma."
- **Budding or cellular tufting:** These are tiny, irregular tightly packed stroma-free clusters of tumor cells which gets detached and float into the lumen of the cyst.
- Mild nuclear atypia.
- Increased mitotic activity.
- Absence of stromal invasion.
- May also show dense and closely packed complex papillae

Malignant:

- **Low-grade serous carcinomas:**
  - It shows variety of architectural patterns.
  - These include single cells and small nets of irregular shape infiltrating the stroma, and micro papillae (rarely macro papillae).
  - The tumor cells are more uniform with mild to moderate nuclear atypia and limited nuclear pleomorphism.
- **High-grade serous carcinomas:** They are distinguished from low-grade carcinoma by:
  - More complex growth patterns with solid masses of tumor cells with slit-like spaces.
  - Widespread invasion or frank effacement of the underlying stroma
  - Tumor cells show marked nuclear atypia, pleomorphism, hyperchromatic nuclei, atypical mitotic figures, and large bizarre form or multinucleation.
  - **Psammoma bodies:** These are concentric calcifications observed in serous tumors. However, they are not specific for neoplasia.

Serous tubal intraepithelial carcinomas: The tumor cells are appear similar to high-grade serous carcinomas but does not show invasion.

RISK FACTORS FOR MALIGNANT SEROUS TUMORS:

- **Parity:** Nulliparous and women with low parity have higher risk.
- Gonadal dysgenesis
- **Genetic factors:** Heritable mutations in tumor suppressor genes BRCA1 and BRCA2
- Family history.
- **Oral contraceptives and sterilization:** During reproductive period reduces the risk.

SPREAD OF OVARIAN SEROUS CARCINOMA

- **Local spread:** Both low- and high-grade carcinomas are commonly associated with ascites. Their spread beyond ovary determines the stage of the disease.
    - Umbilical metastasis ("Sister Joseph's nodule")
    - Contralateral ovary.
    - Abdominal viscera (bowel, liver, spleen)
- **Lymphatic spread:** Para-aortic and pelvic lymph nodes.
- **Hematogenous spread**: Liver and lung.

2. GERM CELL TUMORS OF OVARY

- Germ cell tumors constitute about 20–25% of all ovarian tumors.
- In adults: Most germ cell tumors are benign (mature cystic teratoma, dermoid cyst).
- In children and young adults: Malignant tumors are the most common.
- They are similar to germ cell tumors in the testis.

TERATOMAS

Origin

- All benign ovarian teratomas have a karyotype of 46, XX.
- Teratomas arise from an ovum after the first meiotic division.
- They originate from totipotent cells.
- Teratoma contains mature or immature cells or tissues representative of more than one germ cell layer (at least two) and sometimes all three embryonic layers.
- These cells or tissues are arranged in a helter-skelter fashion (disorganized).

Sites

- **Gonadal**: Ovary and testis.
- **Extragonadal:** Rare and arise from midline embryonic rests, e.g. mediastinum, retroperitoneum.

Classification

- **Mature (benign) teratoma:** It consists of all well- differentiated component parts derived from two or three germ layers (ectoderm, mesoderm and endoderm). Depending on gross feature, they are further categorized as:
    - Cystic (mature cystic teratoma)
    - Solid (mature solid teratoma).
- **Immature (malignant) teratoma:** It consists of less well-differentiated / immature elements.
- **Monodermal** or highly specialized presently called as "monodermal teratoma and somatic-type tumors arising from a dermoid cyst."

A. MATURE TERATOMAS / DERMOID CYST / BENIGN CYSTIC TERATOMA OVARY.

- Most benign ovarian teratomas are cystic and are better known as dermoid cysts (mature cystic teratoma).
- Age group: Usually detected in young women during the active reproductive years with a peak incidence in the third decade.
- Clinically they may be discovered incidentally.

Morphology

- **Gross:** Benign teratomas are bilateral in 10–15% of cases.
- **Appearance:** It is usually unilocular, thick-walled cystic tumor with a smooth, shiny outer surface.
- **Cut section:**
  - Cyst contains yellow or gray, buttery or cheesy sebaceous material with variable amount of hair.
  - Cyst is lined by an opaque, gray-white, wrinkled epidermis.
  - Tooth and areas of calcification are common.
- **Rokitansky nodules/protuberance:**
  - This are one or more foci of rounded nodular structure/s, covered with hair protruding into the lumen of the cyst.
  - These are also known as the mammillary body (tubercle) and dermoid nipple.
  - It shows the greatest variety of tissue types of all three germ cell layers and teeth tend to be located at this site.
- **Microscopy**
  - These tumors mainly show differentiation along ectodermal line.
  - The cyst wall consists of skin (stratified squamous epithelium) with skin appendages (sebaceous glands, hair shafts, and other skin adnexal structures).
  - In most cases, structures from all three germ layers can be identified:
    - ✓ **Ectoderm** (e.g. skin, neural tissue, glia)
    - ✓ **Mesoderm** (e.g. smooth muscle, cartilage, bone, fat)
    - ✓ **Endoderm** (e.g. respiratory tract epithelium, gut, thyroid)

Malignancy in Mature Cystic Teratomas

- One of the mature cellular elements may undergo malignant change in about 1% of the dermoids.
- They tend to occur in older women and include squamous cell carcinoma, thyroid carcinoma, melanoma, basal cell carcinoma, and carcinoid tumor.

Solid Teratoma

- Rarely mature (benign) teratoma can be solid with mature tissues derived from two or three germ layers.
- On gross examination, these tumors are difficult to differentiate from the malignant, immature teratomas.

B. IMMATURE MALIGNANT TERATOMAS

- Immature teratomas are rare tumors composed of variable amounts of tissues, which resemble embryonal and immature fetal tissue.
- **Age:** Mostly in prepubertal adolescents and young women. The mean age is 18 years.

Morphology

- **Gross:** Tumors are bulky with smooth external surface.
- **Cut section:**
  - Predominantly solid and have lobulated and variegated appearance, showing heterogeneous mixture of various tissues.
  - Solid areas may show grossly recognizable immature bone and cartilage, hair, sebaceous material, and calcification.
  - Show areas of necrosis and hemorrhage.
- **Microscopy**
  - Tissues derived from the two or three germ layers.
  - The immature elements include immature neuroepithelium (neuroepithelial rosettes and immature glia), cartilage, bone, muscle, and others.
- **Grading:** It is based on the amount of immature neuroepithelium.

C. MONODERMAL TERATOMAS AND SOMATIC- TYPE TUMORS ARISING FROM A DERMOID CYST

- Specialized teratomas are rare tumors composed entirely of one tissue type.
- The most common among these rare are struma ovarii and carcinoid.
- **Struma Ovarii**
  - It is a cystic lesion composed entirely or predominantly of mature thyroid tissue.
  - These thyroidal neoplasms may hyperfunction and may lead to hyperthyroidism.
- **Ovarian Carcinoid**
  - Probably arise from intestinal epithelium in a teratoma.
  - They may be functional (producing 5-hydroxytryptamine) and result in carcinoid syndrome.
- **Strumal carcinoid:** It is a combination of struma ovarii and carcinoid in the same ovary.

DYSGERMINOMA

- Dysgerminoma is the ovarian counterpart of the seminoma of the testis. Constitutes ~2% of all ovarian cancers and ~50% of malignant germ cell tumors.
- **Age group:** ~75% occur in the second and third decades. Predisposing factors: Gonadal dysgenesis (including pseudohermaphroditism) is one of the risk factors.
- **Functional characteristics:** Most do not show endocrine function.
- **Molecular changes:** Like seminomas, dysgerminomas express transcription factors, namely Oct3, Oct4, and Nanog. These are responsible for pluripotency. They also express the receptor tyrosine kinase KIT.

Morphology:

- **Gross**
  - Usually unilateral (80–90%)
  - Solid, firm, round to oval, encapsulated tumors.
  - **External surface:** Smooth, nodular, convoluted/bosselated
  - **Size:** Varies with a mean diameter 15 cm.
  - **Cut section:** Soft, fleshy and uniformly yellow-white to gray-pink (cream colored).
- **Microscopy**
  - **Pattern**: The tumor cells are arranged in sheets, groups, cords or nests separated by scant fibrous stroma.
  - **Tumor cells:** Resemble primordial germ cells with following characteristics:
  - **Cells:** Individual tumor cells are monotonous (uniform).
  - **Cytoplasm:** It is abundant clear to finely granular glycogen-filled and sometimes fine droplets of fat. The cell membrane is well-defined.
  - **Nuclei:** They are large, regular and centrally placed with fine reticular chromatin and one or more prominent nucleoli.
  - Mitotic figures are usually numerous.
  - **Stroma:** Similar to seminoma, the fibrous stroma is infiltrated with mature lymphocytes and occasional granulomas.

Spread

- **Local spread:** It can spread into peritoneal cavity and is associated with a decreased survival rate.
- **Lymphatic spread:** Through lymphatics, it spreads commonly to the contralateral ovary and retroperitoneal nodes.
- **Blood spread:** Lungs.

Prognosis:

- All dysgerminomas are malignant.
- Dysgerminoma is treated surgically, & 5-year survival for patients with stage I is almost 100%.
- The tumor is highly radiosensitive and also responsive to chemotherapy.

ENDODERMAL SINUS (YOLK SAC) TUMOR

- Endodermal sinus tumor is the second most common malignant tumor of germ cell tumor.
- **Origin:** From differentiation of malignant germ cells along the extraembryonic yolk sac lineage.
- Tumor is rich in D-fetoprotein and D1-antitrypsin.
- **Age:** Children or young women.

Morphology

- **Gross:** Usually unilateral and solid.
- **Microscopy:**
  - **Schiller-Duval body**: This is a glomerulus-like structure composed of a central blood vessel enveloped by germ cells within a space lined by germ cells.
  - **Hyaline droplets:** All show prominent intracellular and extracellular PAS positive hyaline droplets. Some may stain for α-fetoprotein by immunoperoxidase techniques.
- **Tumor Marker:** α-fetoprotein and α1-antitrypsin in tumor cells by immuno-histochemistry α-fetoprotein in the serum

NONGESTATIONAL CHORIOCARCINOMA

- Choriocarcinoma in females may be:
  - **Gestational:** More common of placental origin.
  - **Nongestational:** Germ cell origin can be confirmed only in the prepubertal girl, since after puberty the origin from an ovarian, ectopic pregnancy cannot be ruled out.
- Pure choriocarcinomas of ovary are extremely rare and most coexist with other germ cell tumors.

Morphology

- **Gross:** Solid with large areas of hemorrhage
- **Microscopy:** consists of cytotrophoblast and syncytiotrophoblast without any villi.
- **Behavior:** Aggressive tumors and generally metastasize through the bloodstream to the lungs, liver, bone, and other viscera at the time of diagnosis.

3. SURFACE EPITHELIAL TUMOURS OF OVARY

- They arise from the surface, coelomic or germinal epithelium from the outer surface of the ovary.
- Most primary ovarian tumours arise from Mullerian epithelium.

CLASSIFICATION:

Histological types

- Serous
- Mucinous
- Endometrioid

Biological behavior

- Benign – minimal proliferation
- Borderline – atypical proliferative
- Endometrioid – marked proliferation

	Serous	Mucinous	Endometrioid
Etiology	Accounts for about 30% of all ovarian tumours and 50% of ovarian epithelial tumours 70% - benign or borderline, 30% - malignant.	Accounts for 30% of all ovarian tumours. 80% are benign/borderline and 20% are malignant. Occurs in mid adult life.	Shows tubular glands resembling benign or malignant endometrium.
Risk factors	Nulliparous women with low parity, gonadal dysgenesis in children.	smoking	15-20% cases coexist with endometriosis 15-30% are accompanied by carcinoma of endometrium.
Pathogenesis	Mutations in BRCA1 and BRCA2 tumour suppressor genes.	Mutation in KRAS gene causes mucinous carcinoma	Frequent mutations in PTEN, PIK3CA, KRAS and mismatch DNA repair

ENDOMETRIOID TUMOURS: - Morphology:

Gross: most are solid and exhibit necrotic areas.

Microscopy: glandular pattern resembling endometrial origin.

SEROUS TUMOURS - Morphology:

	Gross	Microscopy
Benign	Cyst wall is smooth glistening without epithelial thickening.	Cyst lined by non-stratified or stratified cuboidal or columnar epithelium. Papillae with fibrovascular core seen Psammoma bodies
Borderline	Cyst cavities filled by increased no. of dense and closely packed cauliflower like papillary projection.	Stratification of epithelium. Budding or cellular tufting. Mild nuclear atypia. Increased mitotic activity. Absence of stromal invasion.
Malignant	Shows mixture of solid and cystic areas with large solid or papillary areas.	Nuclear atypia Pleomorphism Atypical mitotic figures Multi nucleation Psammoma bodies

MUCINOUS TUMOUR - Morphology:

	Gross	Microscopy
Benign	Multilocular ,thin walled cysts Cyst contains sticky semisolid mucinous content	Multiple cysts and gland lined by simple non stratified tall ,non-ciliated, columnar mucinous cells with apical mucin & basal nucleus
Borderline	Similar to mucinous cystadenoma	Stratification of epithelium Budding or cellular tufting Mild nuclear atypia Increased miotic activity Absence of stromal invasion
Malignant	Additional solid regions They show necrosis and haemorrhage.	Epithelial cell stratification and atypia Loss of gland architecture Necrosis Infiltration of serosa is common

- **Spread of serous carcinoma**
  - May spread to peritoneal surfaces and omentum, contralateral ovary, umbilical metastasis and to abdominal viscera.
  - Para aortic and pelvic lymph nodes
  - Hematogenous spread to liver and lung.

- **Spread of mucinous cancer**
  - Peritoneal implant and local invasion into neighbouring structures.
  - Metastasis to distant organs are frequent.

CLINICAL PRESENTATION OF SURFACE EPITHELIAL TUMOURS:

- Benign tumours are mostly asymptomatic.
- Most tumours are non-functional and produce symptoms like
  - Lower abdominal pain and distension.
  - Vaginal bleeding
  - Urinary frequency and dysuria
  - Git symptoms
- Malignant tumours are detected when these spread outside ovary. Symptoms include
  - Progressive weakness, weight loss and cachexia.
  - Massive ascites.
  - Metastasis to opposite ovary, regional lymph nodes and hematogenous spread to liver, lungs and git.

BIOCHEMICAL MARKERS: CA -125, Osteopontin

4. CLASSIFY OVARIAN NEOPLASMS; MOLECULAR PATHOGENESIS AND MORPHOLOGY OF OVARIAN TUMOURS.

CLASSIFICATION

Surface epithelial tumours

- Serous tumours
- Mucinous tumour
- Endometroid tumour
- Brenner tumours

Germ cell tumours

- Teratoma
- Dysgerminoma
- Yolk sac tumour
- Choriocarcinoma
- Embryonal carcinoma

Sex cord stromal tumour

- Fibroma
- Fibrothecoma
- Granulose cell tumour
- Leydig cell tumour

GERM CELL TUMOURS:

Dysgerminoma

- Counterpart of seminomas in males
- Highly radiosensitive
- Usually seen in young women
- Metastasis through lymphatic route
- **Morphology:** Nests of cells, separated by fibrous septae, infiltrated by lymphocytes
- **Tumour markers:** PLAP+, OCT 4+, Nanog+

Embryonal carcinoma

- Arrangement of cell are in cords, sheets or papillary
- Highly pleomorphic

Yolk sac tumour/endodermal sinus tumour

- Centrally, there is a blood vessel surrounded by tumour cells and mimic glomeruloid bodies, also seen in glioblastoma multiforme
- Presence of hyaline globules
- Commonly seen in children
- **Tumour markers:** AFP+, Alpha-1 antitrypsin +

Choriocarcinoma

- Usually occurs in women > 60 Years
- Extreme hemorrhage and necrosis is seen
- **Two kinds of cell:** Cytotrophoblast, Syncytiotrophoblast with multinucleous
- Usually metastasize to lungs-cannonball metastasis
- **Tumour markers:** HCG+ve

Teratoma: Presence of elements from multiple germ layers like hair, bone, cartilage etc.

Mature	Immature	Monodermal
Benign	Malignant	Struma ovary
No primitive element	Presence of primitive element	
If cystic ,dermoid cyst	Poor prognosis	

SEX CORD STROMAL TUMOURS

- Elaborate oestrogen or progesterone
- Can present as precocious puberty

Fibroma

- Histologically composed of well differentiated fibroblasts.
- Benign, unilateral and solid
- Associated with Meigs syndrome i.e. fibroma, ascites and right hydrothorax.

Thecoma

- Composed of spindle shaped cells with lipid droplets (vacuolated cytoplasm)
- Oil red O stain – positive
- Benign, unilateral and solid.

Fibrothecoma: Histologically contains features of both fibroma and thecoma.

Granulosa cell tumour

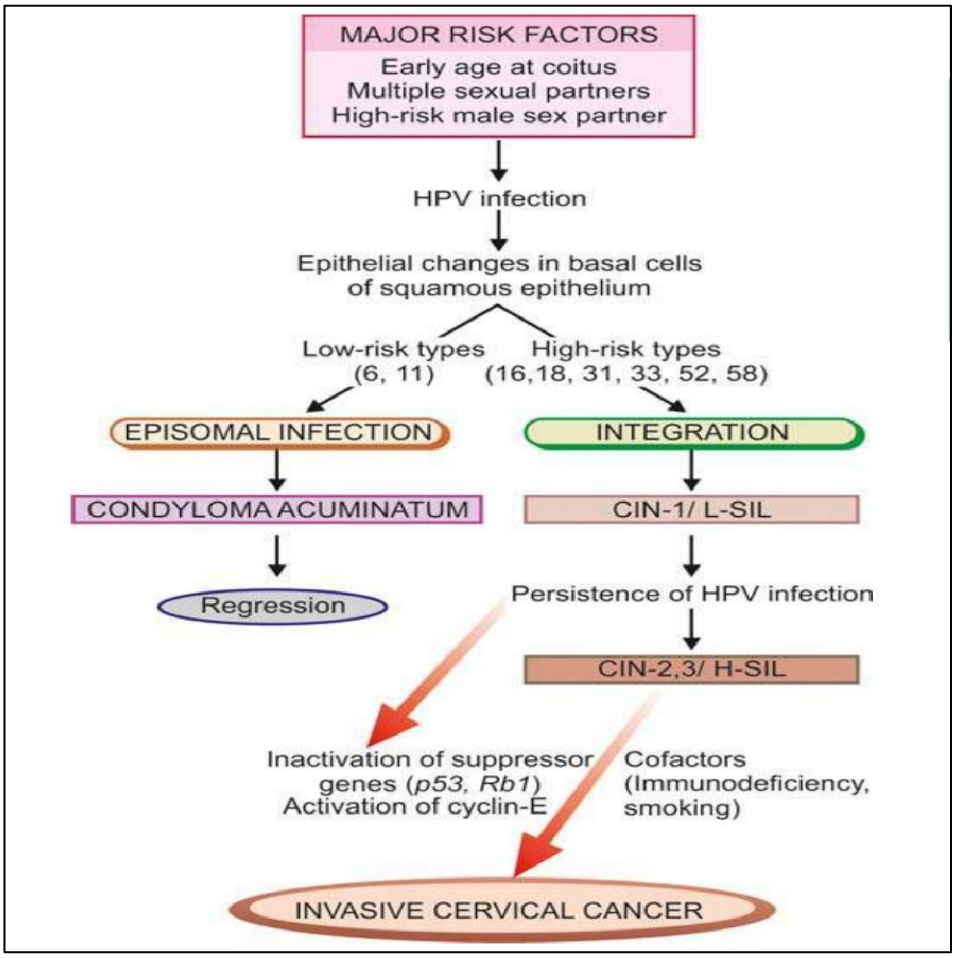
- 5%of ovarian tumours
- Elaborate oestrogen and can cause precocious puberty, endometrial hyperplasia and endometrial carcinoma.
- **Genetic** – FOXL2 gene
- **Marker** – inhibin
- **Gross:** yellowish in colour due to lipids.
- **Microscopy:** cells arranged in cords and sheets
    - Call- Exner bodies
    - Coffee bean nuclei

Leydig cell tumour

- Elaborates testosterone
- Reinke's crystals also seen in normal Leydig cells but its number is significantly increased in tumours.

5. CERVICAL CANCER

- Invasive cervical cancer in about 80% of cases is epidermoid (squamous cell) carcinoma.
- Peak incidence of invasive cervical cancer is in 4th and 6th decade of life.



MORPHOLOGICAL FEATURES

- Invasive cervical carcinoma may present 3 types of patterns:
  - Fungating.
  - Ulcerating.
  - Infiltrating.
- Fungating or exophytic pattern appearing as cauliflower-like growth infiltrating the adjacent vaginal wall is the most common type.
- Characteristically, cervical carcinoma arises from squamocolumnar junction.
- The advanced stage of the disease is characterised by widespread destruction and infiltration into adjacent structures.
- Distant metastases occur in the lungs, liver, bonemarrow, and kidney.

HISTOLOGY

- **Epidermoid (squamous cell) carcinoma**
  - It comprises vast majority of invasive cell carcinoma.
  - The most common pattern(70%) is moderately differentiated non-keratinising large cell type and has better prognosis.
- **Adenocarcinoma -** These may be well-differentiated mucus-secreting adenocarcinoma, or clear cell type containing glycogen but no mucin.
- **Others -** Such as adenosquamous CA, verrucous CA and undifferentiated CA.

CLINICAL FEATURES:

- Cervical cone excision/ hysterectomy with lymph node dissection is done.
- 5yr survival rate is 100% for microinvasive carcinomas & less 50% for tumors beyond pelvis.
- Death due to local tumor invasion (ureteral obstruction, pyelonephritis and uremia)

CLINICAL STAGING

Table 22.4 FIGO clinical staging of carcinoma of the cervix uteri.

Stage	0	Carcinoma *in situ*
Stage	I	Carcinoma strictly confined to the cervix
	IA	Preclinical carcinomas diagnosed only by microscopy
		Maximum size 5 mm deep and 7 mm across measured from the base of epithelium.
	IA1	Stromal invasion of less than 3 mm in depth and 7 mm in horizontal axis (minimally invasive).
	IA2	Stromal invasion of 3 to 5 mm depth and horizontal 7 mm or less (microinvasive).
	IB	Clinical lesion confined to the cervix or preclinical lesions greater than stage IA.
	IB1	Clinical lesions no greater than 4 cm in size.
	IB2	Clinical lesions greater than 4 cm in size.
Stage	II	Carcinoma extends beyond the cervix but has not extended to the pelvic wall. Involvement of the vagina limited to upper two-thirds.
	IIA	No obvious parametrial involvement.
	IIB	Obvious parametrial involvement.
Stage	III	The carcinoma has extended to the pelvic wall. The tumour invades the lower third of vagina.
	IIIA	No extension to the pelvic wall.
	IIIB	Extension to the pelvic wall and/or hydronephrosis or nonfunctioning kidney.
Stage	IV	The carcinoma has extended beyond the true pelvis or has clinically involved the mucosa of the bladder or rectum.
	IVA	Spread of the growth to adjacent organs.
	IVB	Spread to distant organs.

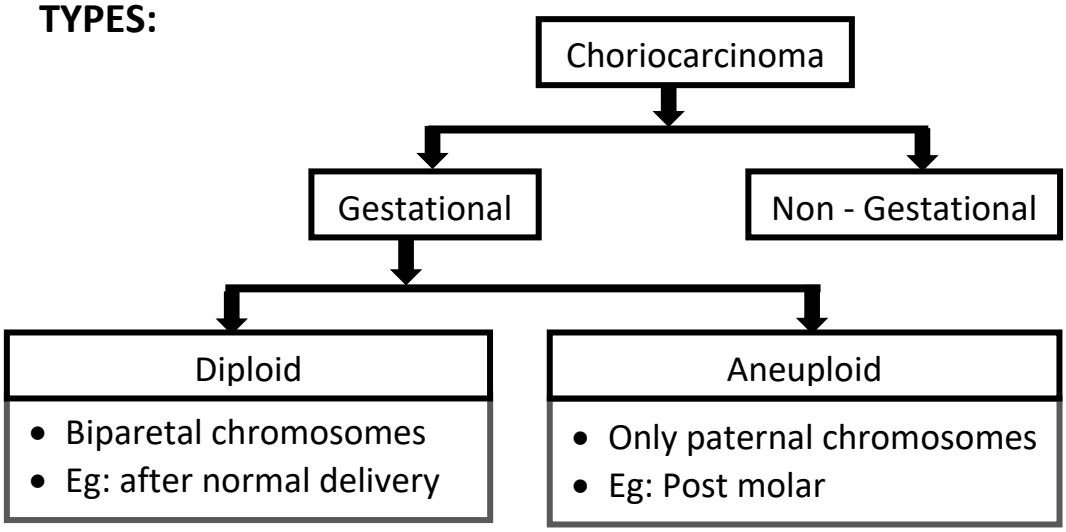
SHORT NOTES

1. CHORIOSTOMA

PATHOGENESIS

- Highly malignant epithelial tumor arising from trophoblastic tissue eg: molar pregnancy or germ cell tumor arising from ovary or testis.
- Excessive expression of paternal genes → excessive proliferation of trophoblastic tissue → gestational trophoblastic disease
- Excessive proliferation of syncytiotrophoblast → ↑ BCG (human chorionic gonadotropin) → ovarian cysts

TYPES:



RISK FACTORS:

- complete molar pregnancy
- advanced maternal age

CLINICAL FEATURES: depends on the organ of metastasis

- **Vagina:** profuse vaginal bleeding, vulvar dark blue papules
- **Lungs:** chest pain, dyspnea, hemoptysis
- **Brain & meninges:** headache, dizziness
- **Liver:** jaundice ,abdominal tenderness

MORPHOLOGY:

Gross: soft, fleshy, yellow white tumor, large pale areas of necrosis & extensive hemorrhage

Microscopy: In gestational choriocarcinoma, proliferating cytotrophoblasts and syncytiotrophoblasts are seen. No chorionic villi is present.

Diagnosis: CT, MRI, X RAY, Pelvic ultrasound

Lab results: ↑ serum beta HCG, CBC – anemia

Complications: In highly vascularized tumor, profuse bleeding hematogenous metastasis to other organs (eg: lung, brain, liver)

2. BRENNERS TUMOR:

- They are transitional cell tumors with neoplastic epithelial cells resembling urothelium.
- They are usually benign
- 10% of ovarian epithelial tumors.
- detected incidentally
- Even in large size behaves in a benign fashion.
- low grade carcinoma (type 1) → malignant Brenner's tumors
- high grade carcinoma (type 2) → uncommon transitional cell carcinoma
- neoplasms may have hormonal activity
- benign Brenner's tumor with malignant tumor cells become malignant Brenner's tumor

RISK FACTORS: ovarian cancer, BRCA mutations

MORPHOLOGY:

- solid or cystic neoplasm
- unilateral
- size → small lesions to massive tumors
- Fibrous stroma → contain demarcated nests of epithelial cells with mucinous glands in their centre. (sometimes contain plump fibroblast resembling theca cells)

CLINICAL FEATURES:

- lower abdominal pain
- abdominal enlargement
- GIT complaints
- urinary frequency
- dysuria
- pelvic pressure
- In malignant conditons ,
    - progressive weakness
    - weight loss
    - cachexia
    - massive ascites → ascites fluid filled with exfoliated tumor cells
    - peritoneal pattern of spread is seen → serosal surface has nodules of tumor
    - metastasis from midline to opposite ovary is seen

DIAGNOSIS: biochemical tests for tumor antigens or products, serum marker CA -125

TREATMENT: salpingo oophrectomy

3. BENIGN CYCTIC TERATOMA

Most benign teratomas are cystic and are often referred to as dermoid cysts, since they are almost always lined by skin - like structures.

OCCURENCE:

- Usually found in young women during the active reproductive years.
- Discovered incidentally, but are occasionally associated with clinically important paraneoplastic syndromes, such as inflammatory limbic encephalitis, which may remit upon removal of the tumor.

MORPHOLOGY:

- bilateral in 10% to 15% of cases.
- unilocular cysts containing hair and sebaceous material
- Sectioning - reveals a thin wall lined by an opaque, gray-white, wrinkled epidermis, frequently with protruding hair shafts. Within the wall, it is common to find grossly evident tooth structures and areas of calcification.
- Microscopically - the cyst wall is composed of stratified squamous epithelium with underlying sebaceous glands, hair shafts, and other skin adnexal structures.
    - In most cases tissues from other germ layers can be identified, such as cartilage, bone, thyroid, and neural tissue.
    - Dermoid cysts are sometimes incorporated within the wall of a mucinous cystadenoma.
- About 1% of the dermoids undergo malignant transformation, most commonly to squamous cell carcinoma, but also to other cancers as well (e.g., thyroid carcinoma, melanoma).
- In rare instances a benign teratoma is solid and composed entirely of benign-looking heterogeneous collections of tissues and organized structures derived from all three germ layers.
- These tumors have the same histogenetic origin as dermoid cysts but lack preponderant differentiation into ectodermal derivatives.
- These neoplasms may be difficult to distinguish, on gross inspection, from malignant immature teratomas.

GENETICS:

The karyotype of almost all benign ovarian teratomas is 46,XX. Genetic analyses indicate that the majority of teratomas arise from an ovum after the first meiotic division, while a minority arises before the first division.

4. ENDOMETRIOSIS

Endometriosis is defined by the presence of "ectopic" endometrial tissue at a site outside of the uterus.

SITES:

- Ovary - most common;
- Uterine ligament
- Retro uterine pouch
- Bladder / bowel
- Retro vaginal septum
- pelvic peritoneum,
- appendix,
- Laparotomy scars.
- Mucosa of fallopian tube, cervix, vagina

MORPHOLOGY:

- Endometriotic lesions bleed periodically in response to both extrinsic cyclic (ovarian) and intrinsic hormonal stimulation.
- This bleeding produces nodules with a red-blue to yellow-brown appearance on or just beneath the mucosal and/or serosal surfaces at sites of involvement.
- When lesions are extensive, organizing hemorrhage causes extensive fibrous adhesions between tubes, ovaries, and other structures and obliterates the pouch of Douglas.
- The ovaries may become markedly distorted by large cystic masses (3 to 5 cm in diameter) filled with brown fluid resulting from previous hemorrhage; referred to clinically as chocolate cysts or endometriomas.
- Aggressive forms of endometriosis can invade tissues and cause fibrosis and subsequent adhesions.
- The histologic diagnosis of endometriosis is usually straight- forward but may be difficult in long-standing cases in which the endometrial tissue is obscured by secondary fibrosis.
- The diagnosis is readily made when both endometrial glands and stroma are present with or without the presence of hemosiderin.
- In rare cases only stroma is identified. If only glands are present, other diagnoses with different clinical ramifications, such as endosalpingiosis, must be considered.
- Atypical endometriosis, the likely precursor to endometriosis-related ovarian carcinoma, has two morphologic appearances:
  - One consists of cytologic atypia of the epithelium lining the endometriotic cyst without major architectural changes.
  - The second is marked by glandular crowding due to excessive epithelial proliferation, often associated with cytologic atypia, producing an appearance that resembles complex atypical endometrial hyperplasia.

PATHOGENESIS:

- The regurgitation theory proposes that endometrial tissue implants at ectopic sites via retrograde flow of menstrual endometrium.
- The benign metastases theory holds that endometrial tissue from the uterus can "spread" to distant sites (e.g., bone, lung, and brain) via blood vessels and lymphatic channels.
- The metaplastic theory suggests that endometrium arises directly from coelomic epithelium (mesothelium of pelvis or abdomen), from which the müllerian ducts and ultimately the endometrium itself originate during embryonic development.
- The extrauterine stem/progenitor cell theory is a recent idea that proposes that stem/progenitor cells from the bone marrow differentiate into endometrial tissue.

MOLECULAR ANALYSIS:

- Release of proinflammatory and other factors, including PGE2, IL-1β, TNFα, IL-6 and -8, NGF, VEGF, MCP-1, MMPs, and TIMPs.
- Increased estrogen production by endometriotic stromal cells, due in large part to high levels of the key steroidogenic enzyme aromatase, which is absent in normal endometrial stroma.
- Shared mutations in specific genes (PTEN and ARID1A) in endometriotic cysts, atypical endometriosis and associated carcinomas.

CLINICAL FEATURES:

- Severe dysmenorrhea
- Dyspareunia (pain with intercourse)
- Pelvic pain due to the intrapelvic bleedIng and periuterine adhesions.
- **Ovary -** most common; chocolate cyst; reduces fertility.
- **Uterine ligament -** causes pelvic pain and dysmenorrhea.
- **Retro uterine pouch** - pain on defecation.
- **Bladder / bowel** - pain on urination / defecation / abdominal pain.
- **Mucosa of fallopian tube** - reduces fertility.
- Menstrual irregularities are common
- Infertility is the presenting complaint in 30% to 40% of women.
- Associated condition: Adenomyosis, the presence of endometrial tissue within the uterine wall (myometrium)

TREATMENT: Gonadotropin-releasing hormone (Gn-RH) agonists and antagonists, Progestin therapy, Aromatase inhibitors.

5. MALIGNANT TUMORS OF ENDOMETRIUM

Endometrial carcinoma is the most common invasive cancer of the female genital tract.

TYPE 1 (ENDOMETRIOD CANCER)

- **Age**: 55-65 years
- **Clinical setting**
  - Unopposed estrogen exposure
  - Obesity
  - Hypertension
  - Diabetes Mellitus
- **Genetics**
  - PTEN mutation
  - Indolent in nature
  - Lymph node spread
  - Better prognosis
- **Morphology**
  - Contain foci of squamous differentiation.
  - Squamous elements may be histologically benign appearing when they are associated with well differentiated adenocarcinomas.
  - Not found in moderate and poorly differentiated forms.
- **Histology**
  - **Grade I** - well differentiated; composed almost entirely of well-formed glands
  - **Grade II** - moderately differentiated showing well-formed glands mixed with areas composed of solid sheets of cells, which make up 50% or less of the tumor.
  - **Grade III -** poorly differentiated characterized by greater

TYPE 2 (PAPILLARY SEROUS CANCER)

- **Age**: 65-75 years
- **Clinical setting**
  - Atropy
  - Thin physique
- **Genetics**
  - P53 mutation
  - Aggressive in nature
  - Intra peritoneal spread
  - Poor prognosis
- **Morphology:** consists of malignant cells identical to those of serous carcinoma that are confined to the epithelial surfaces.
- **Histology** All of the tumors are classified as grade 3 irrespective of histologic pattern.

MALIGNANT MIXED MÜLLERIAN TUMOR:

- MMMTs (also referred to as carcinosarcomas) are endometrial adenocarcinomas with a malignant mesenchymal component.
- **Morphology:** Bulky and polypoid, and may protrude through the cervical os.
- **Histology :**
  - Consist of adenocarcinoma (endometrioid, serous, or clear cell) mixed with the malignant mesenchymal (sarcomatous) elements.
  - Alternatively, the tumor may contain two distinct and separate epithelial and mesenchymal components.
  - Sarcomatous components may also mimic extrauterine tissues (e.g., striated muscle, cartilage, adipose tissue, and bone).
  - Metastases usually contain only epithelial components.
- **Clinical features :** Post-menopausal bleeding

6. CHORIOCARCINOMA

- Gestational choriocarcinoma is an uncommon highly invasive malignant neoplasm of trophoblastic cells derived from a previous normal or abnormal pregnancy.
- Choriocarcinoma metastasizes widely via hematogenous spread.
- Most common sites for metastases in the order of decreasing frequency are: Liver (50%), vagina, brain, liver, bone and kidney.
- Very rarely, a nongestaional-choriocarcinoma may develop from ovarian germ cells.
- **Morphology & histopathology:**
  - Soft, fleshy, yellow-white tumor with large areas of necrosis and hemorrhage.
  - Chorionic villi – Absent.
  - Contains rapidly proliferating synctiotrophoblasts and cytotrophoblasts.
  - Often invades uterine wall and penetrates blood vessels.
- **Clinical features:**
  - Irregular vaginal spotting (bleeding) of bloody, brown fluid.
  - Typically appears after a normal pregnancy or after a miscarriage.
  - HCG levels are usually elevated.
- **Treatment:**
  - Treatment for gestational choriocarcinoma involves-Evacuation and Chemotherapy.
  - Response to chemotherapy is high (almost 100%).
  - Cured patients can have normal pregnancies and deliveries.
  - Nongestational choriocarcinoma are somewhat resistant to therapy.

7. CERVICAL INTRAEPITHELIAL NEOPLASIA (CIN)

- CIN is a precancerous condition for cervical carcinoma characterized with abnormal growth of squamous epithelium of cervix.
- It is most commonly seen at squamo-columnar junction of cervix.
- CIN is also known as Squamous Intraepithelial Lesions (SIL) / cervical dysplasia.
- HPV is necessary for development of CIN, but not all HPV infections leads to CIN.
- HPV-16 is the most common HPV type for development of these lesions.
- Typically, CIN resolves on its own.

CLASSIFICATION:

Oldest Classification	CIN	Latest
Mild dysplasia	CIN 1	LSIL
Moderate dysplasia	CIN 2	HSIL
Severe dysplasia / Carcinoma in situ	CIN 3	HSIL

MORPHOLOGY & HISTO-PATHOLOGY:

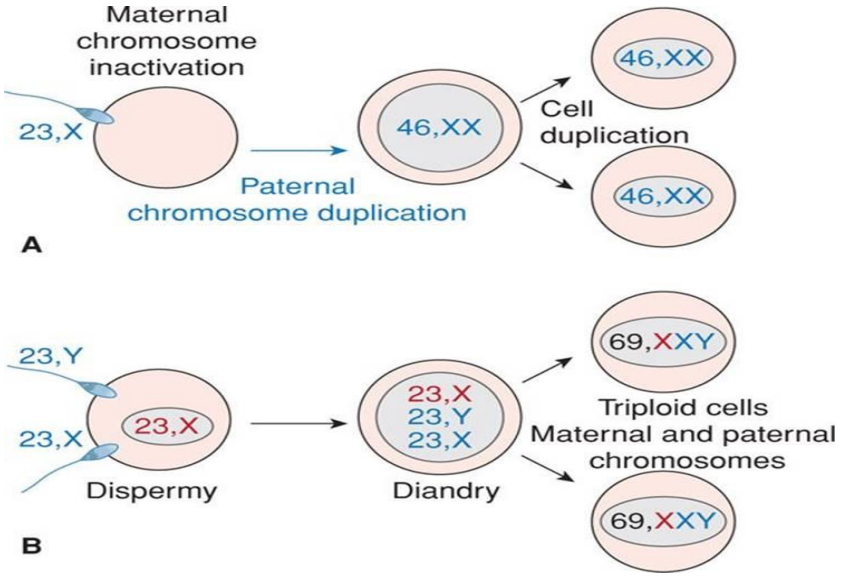
- Nuclear Atypia – nuclear enlargement, hyperchromasia (dark staining).
- Cytoplasmic 'halos' are seen as a cytopathic change of ER membranes because of E5 protein produced by HPV.
- Nuclear alterations with perinuclear changes is known as –KOILOCYTIC ATYPIA.
- Grading of SIL is done based on expansion of immature squamous cells from basal layer.
- LSIL-immature squamous cells are confined to lower one-third of lesion.
- HSIL-immature squamous cells expand to upper two-thirds of lesion.
- HPV E6 and E7 proteins prevent cell cycle arrest. As a result, surface cells express markers of actively dividing cells like Ki-67.
- Ki-67 and p16 staining are useful for diagnosis of HPV infection and SIL.

LSIL	HSIL
Mild dysplasia.	Moderate/severe dysplasia.
Lower 1/3 rd. of epithelium.	Expand to upper 2/3rds of epithelium.
High rate of viral replication.	Lower rate of replication, as compared to LSIL.
Mild alteration in growth of host cells.	Deregulation of cell cycle leads to ↑ cell proliferation
60% cases regress spontaneously.	HSILS are at higher risk for carcinoma.
10% cases progress to HSIL.	10% cases progress to carcinoma, Within 2-10 yrs.

8. HYDATIDIFORM MOLE

- Hydatidiform mole is a benign neoplasia of chorionic villi with malignant potential.
- Moles are characterizes histologically by cystic swellings of the chorionic villi, with trophoblastic proliferation.
- Moles are associated with increased risk of choriocarcinoma and invasive mole.

- **Complete mole:**
  - Complete mole results from fertilization of an inactivated ovum i.e. ovum without female chromosomes.
  - MOLAR PREGANANCY usually refers to the complete mole.
  - Molar pregnancy can develop at any age, but risk is higher at two ends of reproductive age (in teenagers and >40 yrs.).
  - 90% cases- 46, XX karyotype – duplication of chromosomes of one sperm (ANDROGENESIS).
  - 10% cases- 46, XX / 46, XY –empty egg fertilizes with two sperms (dispermy).



- **Partial mole:**
  - Partial mole results from fertilization of egg with two sperms. Thus, these moles have a triploid/tetraploid karyotype (69, XXY / 92, XXXY).
  - Partial moles have a lower risk for development of choriocarcinoma.

- **Invasive/persistent mole:**
  - If the mole penetrates the uterine wall then it is known as an invasive mole.
  - This tumor is locally destructive and may invade entire parametrial tissue (uterine wall) and blood vessels.
  - Hydropic villi, thus penetrated may embolize to distant locations like lungs and brain, but do not grow as true metastases.
  - This tumor may cause excessive uterine wall rupture and may necessitate hysterectomy (surgical removal of uterus).

MORPHOLOGY & HISTOPATHOLOGY:

- Delicate, thin-walled, grape like vesicular cysts containing edematous villi.
- Enlarged, fluid-filled cysts are lined by trophoblasts.
- In complete mole – entire villi are covered by trophoblastic proliferation.
- In partial mole – only a fraction i.e. focal hyperplasia is seen.

CLINICAL FEATURES:

- Abnormal vaginal bleeding, expulsion of grapelike vesicular tissues.
- In complete mole, Elevated β-HCG levels compared to of levels in normal pregnancy of same gestational age.
- Most moles are successfully removed by suction, evacuation and curettage.
- Prophylactic chemotherapy is indicated if β-HCG do not fall within 3 months after removal of mole (indicates presence of invasive mole).
- 2.5% of complete moles give rise to gestational choriocarcinoma.

9. CERVICAL CANCER- LAB DIAGNOSIS

Cytological cervical cancer screening helps in preventing most cancers arising from precursor lesion.

PAPANICOLAOU / PAP SMEAR TEST:

- A Pap smear is a screening procedure for cervical cancer.
- It tests for the presence of precancerous or cancerous cells on the cervix.
- Entire spectrum of precancerous condition can be assessed by Pap test
- It is recommend that women get regular Pap smears every three years starting at age 21 or within 3 years of onset of sexual activity.
- After age 30, women who have normal cytology and negative HPV test are to be screened for every 5 years.
- Scraping of abnormal cells using a spatula or brush → Fixation and staining using Papanicolaou method → Smears are screened microscopically

HPV DNA TESTING:

- HPV testing has a higher sensitivity but lower specificity as compared to Pap test, because not all HPV infections leads to cervical cancer.
- If the result of a pap test is abnormal, a colposcopic examination is performed to identify the lesion.
- Management of LSIL involves local ablation and HSILs are treated with cervical conization/excision.

10. UTERINE LEIOMYOMA

- Uterine leiomyoma (commonly called fibroids) is the most common tumor in women.
- They are benign smooth muscle neoplasms that may occur singly, but more often are multiple.
- **Genetics :**
  - 40% have a simple chromosomal abnormality.
  - Several cytogenetic subgroups are recognized, including tumors with rearrangements of chromosomes 12q14 and 6p involving the HMGIC and HMGIY genes.
  - Mutations in the MED12 gene have been identified in up to 70% of uterine leiomyomas.

MORPHOLOGY:

- Leiomyomas are sharply circumscribed, discrete, round, firm, and gray-white tumors varying in size from small, barely visible nodules to massive tumors that fill the pelvis (rarely found within the myometrium of the corpus).
- Infrequently do they involve the uterine ligaments, lower uterine segment, or cervix
- Large tumors may develop areas of yellow-brown to red softening.
- They can occur within the myometrium (intramural), just beneath the endometrium (submucosal) or beneath the serosa (subserosal).

HISTOLOGY:

- Characteristic whorled pattern of smooth muscle bundles
- Leiomyomas are typically composed of bundles of smooth muscle cells that resemble the uninvolved myometrium.
- Usually, the individual muscle cells are uniform in size and shape and have the characteristic oval nucleus and long, slender bipolar cytoplasmic processes.
- Mitotic figures are scarce.
- Benign variants of leiomyoma include atypical or bizarre (symplastic) tumors with nuclear atypia and giant cells, and cellular leiomyomas. Both have a low mitotic index, helping to distinguish these benign tumors from leiomyosarcomas.
- Benign metastasizing leiomyoma, is a uterine leiomyoma that extends into vessels and spreads hematogenously to other sites, most commonly the lung.
- Disseminated peritoneal leiomyomatosis, presents as multiple small peritoneal nodules.
- But both are considered benign despite their unusual behavior.

CLINICAL FEATURES:

- Abnormal bleeding
- Urinary frequency due to compression of the bladder
- Sudden pain from infarction of a large or pedunculated tumor,
- Impaired fertility.
- Myomas in pregnant women increase the frequency of spontaneous abortion, fetal malpresentation, uterine inertia (failure to contract with sufficient force), and post-partum hemorrhage.
- Malignant transformation to leiomyosarcoma, is extremely rare.

TREATMENT: Surgery the gold standard for the treatment (hysterectomy / myomectomy)

11. KRUKENBERG TUMOR:

- **Pathogenesis :**
  - Ovarian cancer metastasized from another primary site (usually from GIT / breast)
  - Spreads mainly by lymphatics
  - Involves both the ovaries
- **Clinical features :**
  - pelvic / abdominal pain
  - vaginal bleeding
  - bloating
  - ascites
  - dyspareurnia
- **Morphology:** mucin secreting signet ring cells are present.
- **Diagnosis:** CT, MRI, biopsy, laparotomy
- **Treatment :** surgery ,chemotherapy

SHORT ANSWERS:

1. SCILLER DUVEL BODIES

- They are a characteristic histological feature yolk sac/ endodermal sinus tumor of malignant germ cell origin
- It is a glomerulus shaped structure; composed of central blood vessel surrounded by tumor cells

2. CALL EXNER BODIES

- Characteristic feature of granulosa cell tumors consisting of large solid cystic masses
- The bodies are small, gland like structure filled with acidophilic material

3. GROSS FEATURES OF MATURE CYSTIC TERATOMA

- Benign teratomas which may present bilaterally
- They are seen as unilocular cysts containing hair and sebaceous material
- On sectioning, a thin wall lined by an opaque, greyish wrinkled epidermis with protruding hair shafts and tooth structures (calcification areas) are seen.

4. HISTOLOGIC FEATURES OF DYSGERMINOMAS

- Dysgerminomas are ovarian counterpart of testicular seminomas and are mostly unilateral
- On cut section, they have yellow – pink appearance that are soft and fleshy.
- They have large vesicular cells with clear cytoplasm; tumor cells grow in sheets or cords separated by scant fibrous stroma infiltrated by mature lymphocytes and granuloma

5. CLASSIFICATION OF GESTATIONAL TROPHOBLASTIC DISEASE:

- It is a spectrum of tumor and tumor like condition due to proliferation of placental tissue either villous and trophoblast
- **Classification:**
  - Hydatidiform mole- Complete / Partial
  - Invasive mole
  - Choriocarcinoma
  - PSTT [Placental Site Trophoblastic Tumour]

6. ADENOMYOSIS:

- A clinical feature of endometriosis
- Presence of endometrial tissue within myometrium
- Continuous with endometrium as a downgrowth
- **Microscopy:** Irregular nest of endometrial stroma with or without gland within myometrium seperated feom basalis by 2-3 mm
- **Clinical symptoms:**
  - Menometrorrhagia [heavy irregular mensus]
  - Cloicky dysmenorrhoea
  - Dyspareunia
  - Pelvic pain in pre menstural period

7. DERMOID CYST OF OVARY

- Most benign tumors are cystic in nature so they are called dermoid cysts as they are lined by skin structures
- They have structures of different germ layer origins
- They can also undergo malignant transformation (squamous cell carcinoma)

8. MICROSCOPIC FEATURE OF LEIOMYOMA:

- Has bundles of smooth muscles
- Individual muscle cell: Normal uniform size, shape, oval nucleus, slender bipolar cytoplasmic processes
- Scarce mitotic figures

9. ALPHA FETOPROTEIN:

- A serum marker for tumor
- Commonly seen in pregnancy patients
- Related to embryological development for diagnosing germ cell tumors and liver tumors like hepatocellular carcinoma and hepatoblastomas
- Normal level: <10ng/mL
- If >500ng /mL it indicates risk of several tumors

UPDATES

1. RELEASE OF FACTORS IN CHRONIC ENDOMETRIOSIS:

- Release of pro inflammatory and angiogenic factors, including PGE2, IL-1β, TNFα, IL-6 and IL-8, NGF, VEGF, MCP-1, MMPs, and TIMPs.
- Some of these factors are released by associated macrophages, which are recruited to endometriotic implants by pro inflammatory factors.
- Like metastatic tumors, the ability of endometriotic implants to survive and grow is dependent on angiogenesis, which is mediated by typical pro angiogenic factors such as VEGF.
- Similarly, the ability to implant requires remodeling of the extracellular matrix, which is carried out by factors such as matrix metallo proteases. These factors may also contribute to avoidance of immune clearance.

2. MUTATIONS IN CHRONIC ENDOMETRIOSIS

- Mutations such as KRAS, PIK3CA, PPP2R1A, and ARID1A, have been identified in the epithelial cells of deeply infiltrating endometriosis.
- They contribute to the locally aggressive behavior of deeply infiltrating endometriosis.

3. MOLECULAR SUBTYPES OF ENDOMETRIAL CARCINOMA

- More recently, genomic sequencing of endometrial carcinoma has revealed four major molecular subtypes among endometriod and serous carcinomas, the most common morphologies of type 1 and type 2 tumors, respectively.
- The salient molecular features of these four subtypes are as follows:
  - Ultramutated/POLE tumors, which are defined by the presence of mutations in DNA polymerase ε (POLE) that produce an exceptionally high burden of somatic mutations, the large majority of which are passenger mutations.
  - Hypermutated/MSI (microsatellite instability) tumors, defined by mutations in or epigenetic silencing of mismatch repair genes, also leading to genomic instability and a high burden of somatic mutations.
  - Copy number low/MSS (microsatellite stable) tumors, a common subtype also associated with endometrioid morphology that is frequently associated with mutations that up regulate signaling through the PI3K/AKT pathway.
  - Copy number high/serous-like tumors, aggressive tumors with serous or high-grade endometrioid morphology that are often associated with TP53 mutations and numerous genomic copy number variants.

4. ONE LINERS:

- HPV vaccines provide excellent protection against infection by low-risk HPV and genital warts.
- IA1—Stromal invasion no deeper than 3 mm and no wider than 7 mm (so-called superficially invasive squamous cell carcinoma).
- Retrograde menstruation through the fallopian tubes is common, and the regurgitation theory provides a plausible explanation for the origin of ectopic endometrial tissue in the peritoneal cavity, which constitutes the vast majority of cases.
- Retinoic acid produced in the stromal cells promotes epithelial cell survival via paracrine signaling
- A small subset (less than 10%) of endometrial carcinomas have mutations that disrupt the proofreading function of DNA polymerase ε, encoded by the POLE gene. These tumors have a remarkably high burden of somatic point mutations, possibly the highest of any human cancer.
- Tumors with polymerase ε mutations or defects in DNA mismatch repair are frequently associated with large numbers of infiltrating T cells. It is believed that these cells represent an ineffective host response to neo antigens created by mutations that lead to tumor-specific amino acid substitutions in proteins.

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